GATE-AII BRANCHES
Engineering Mathematics



PROBABILITY AND STATISTICS

Lecture No.- 17



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Recap of Previous Lecture



Topic Question based on uniform distribution / Gaussian Distribution

Topics to be Covered



Properties of expected value.

Topic Poisson distribution

Topic Questions based on poisson distribution

Properties of expected value:

(A) If X and Y Are Independent Random variables:

$$E[X+Y] = E[X] + E[Y]$$

Joint expected value = Individual random variable

Different

$E(X) = 0.5$

X: Tossing A coin

X = No. of HEADS / No. of tails

X	0(T)	1(H)
P(X=x)	$\frac{1}{2}$	$\frac{1}{2}$

$$E[X] = 0 \times \frac{1}{2} + 1 \times \frac{1}{2} = 0.5$$

Y: Throwing A Die $\equiv$

Y = No. of dots

$E[Y] = 3.5$

Y	1	2	3	4	5	6
P(Y=y)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E[Y] = (1+2+3+4*5+6)\frac{1}{6} = 3.5$$

$X+Y = E[X+Y]$

$X+Y$	1	2	3	4	5	6	7
$P(X+Y = x_i+y_j)$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{2}{12}$	$\frac{1}{12}$

$$E[X+Y] = \left(1\times\frac{1}{12} + 7\times\frac{1}{12}\right) + (2+3+4+5+6)\frac{2}{12}$$

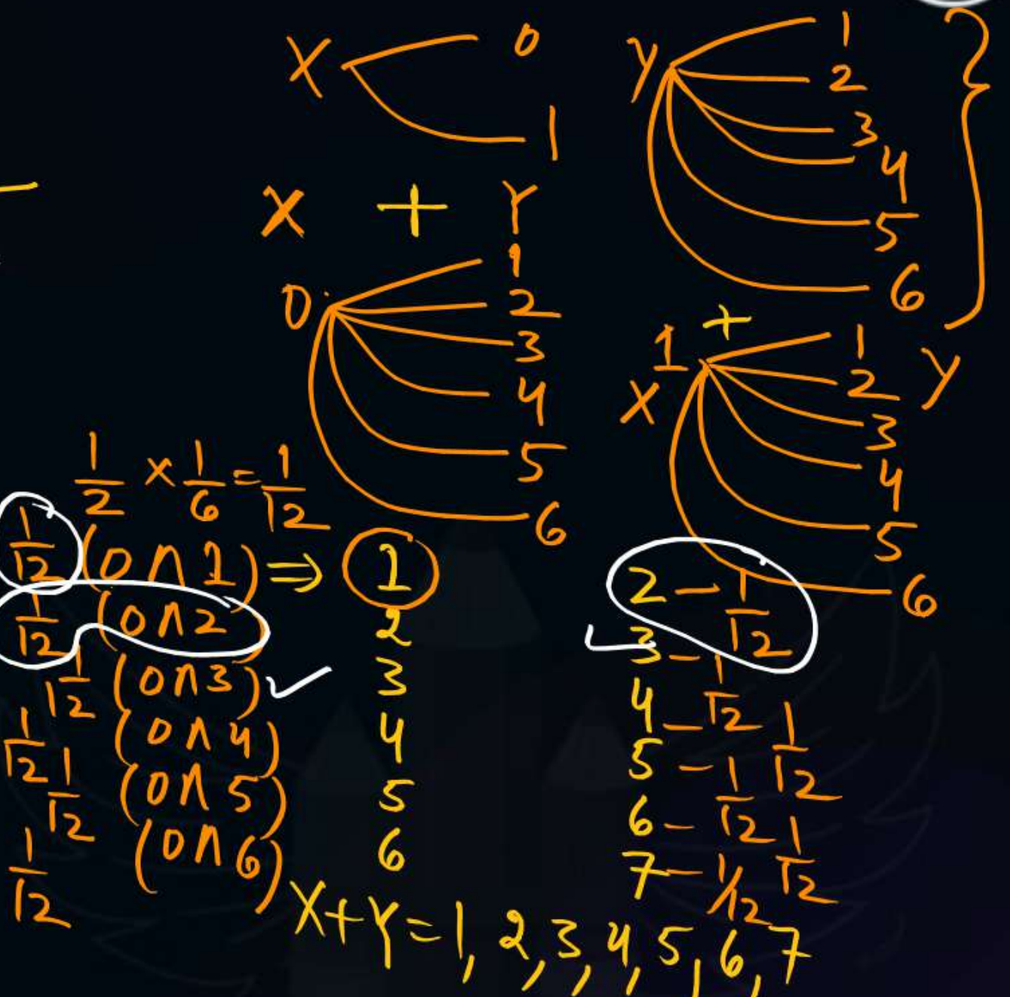
$$= \frac{8}{12} + \frac{40}{12} = \frac{48}{12}$$

$$\boxed{E[X+Y] = 4}$$

$E[X] = 0.5 \quad E[Y] = 3.5$

Verified

$P(X \cap Y) = P(X)P(Y)$



$\frac{1}{2}\times\frac{1}{6} = \frac{1}{12}$

$\frac{1}{12}$ $(0 \cap 1) \Rightarrow 1$

$\frac{1}{12}$ $(0 \cap 2)$

$\frac{1}{12}$ $(0 \cap 3)$

$\frac{1}{12}$ $(0 \cap 4)$

$\frac{1}{12}$ $(0 \cap 5)$

$\frac{1}{12}$ $(0 \cap 6)$

$X+Y = 1, 2, 3, 4, 5, 6, 7$



Superposition Principle

If $x_1, x_2, x_3 -- x_n$ Are 'n' Independent random variable

$$E[x_1+x_2+x_3+--x_n] = E[x_1]+E[x_2]+--+E[x_n]$$

$$E[x \pm y] = E[x] \pm E[y]$$

X_1 = Tossing A coin

X_2 = Throwing A Die

X_3 = Picking A ball

(B) $E\left[\frac{1}{x}\right] \neq \frac{1}{E[x]}$

(C) $E[x^2] \neq [E[x]]^2$

D) $E[ax+b] = aE[x]+b$

E) $E[ax] = aE[x]$.

(F) $\text{var}(-x) = (-1)^2 \text{var}(x)$

G) $\text{var}(ax+b) = a^2 \text{var}(x)$

H) $\text{var}(ax+by) = a^2 \text{var}(x) + b^2 \text{var}(y)$

I) $E\left[\frac{1}{\log x}\right] \neq \frac{1}{E[\log x]}$

Poisson Distribution:

PER / Average
= Poisson Distribution
= (Discrete random Variable)

→ Arrival – discrete

Average Number of success Per unit time

A) Telephone calls Per unit 60 seconds
B) Average No. of customer Arrive between 5 PM to 6 PM
C) Average No. of customer/vehicle arriving 1 PM to 2 PM.
D) Average No. of customer claiming Insurance.
E) average No. of Errors in a book.
F) Average No. of STUDENTS asking doubts

- Bernoulli – Single
- binomial – No. of Trials measured
- Poisson – Average No. of success.

Average No. of success

$$P(X=r) \Rightarrow \frac{e^{-\lambda}\lambda^{r}}{r!}$$

$P(\lambda)$

λ = mean

r = No. of success

100 Hrs — 30 days — $3\frac{1}{2}$ Hrs.

Average No. of success

Where λ = mean of the Random variable

λ = No. of Trials × prob.

$$\lambda = np$$

mean = binomial distribution

(Average No. of success)

Statistical Averages:

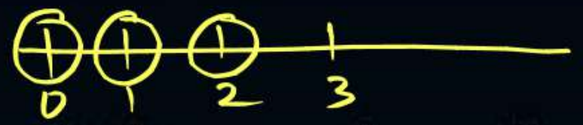
$E[X]$ = mean = $np = \lambda$

$V(X)$ = variance = $np = \lambda$

mean And variance are equal.

Standard deviation = $\sqrt{\lambda}$

Topic : Probability and statistics



Poisson
5 PM to 6 PM
$\lambda = 3$

$$P(X=r) = \frac{e^{-\lambda}\lambda^{r}}{r!}$$

#Q. Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and P has a poission's distribution with the mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

$$P(X=0) = \frac{e^{-3}(3)^{0}}{0!} = e^{-3}$$

$$P(X=1) = \frac{e^{-3}(3)^{1}}{1!} = 3e^{-3}$$

$$P(X=2) = \frac{e^{-3}(3)^{2}}{2!} = \frac{9}{2}e^{-3}$$

$$P(X<3) = P(X=0) + P(X=1) + P(X=2)$$

$$= e^{-3} + 3e^{-3} + \frac{9}{2}e^{-3}$$

$$= \frac{2e^{-3} + 6e^{-3} + 9e^{-3}}{2}$$

$$= \frac{17}{2e^{3}}$$

A. $\frac{8}{2e^3}$

B. $\frac{8}{(2e^3)}$

C. $\frac{17}{2e^3}$

D. $\frac{26}{(2e^3)}$



Topic : Probability and statistics

GATE

#Q. The number of accidents occuring in a plant in a month follows Poisson distribution with mean as 5.2. The probability of occurence of less than 2 accidents in the plant during a randomly selected month is

$$P(X<2) = e^{-5.2} + e^{-5.2}(5.2)^1$$
$$= 0.034$$

A 0.029

B 0.034

C 0.039

D 0.044

$\lambda = 5.2$

$P(X<2) = P(X=0) + P(X=1)$

$P(X=r) = \frac{e^{-\lambda}\lambda^r}{r!}$

$P(X=0) = \frac{e^{-5.2}(5.2)^0}{0!}$

$P(X=1) = \frac{e^{-5.2}(5.2)^1}{1!}$



Topic : Probability and statistics

$e = 2.732$

#Q. The second moment of a poisson-distributed random variables is 2. The mean of the random variable _____ is

$E[X]$ = First moment
$E[X^2]$ = second ''
$E[X^3]$ = Thirer moment
$E[X^n]$ = n^{th} moment

$E[X^2] = 2$ mean $= \lambda$

$P(\lambda) \rightarrow$ var X $= \lambda$, $E(X) = \lambda$

$V(X) = E[X^2] - [E[X]]^2$

$\lambda = 2 - \lambda^2$

$\lambda^2 + \lambda - 2 = 0$

$\lambda^2 + 2\lambda - \lambda - 2 = 0$

$\lambda(\lambda+2) - 1(\lambda+2) = 0$

$\lambda = -2, \lambda = 1$

X $\lambda = -2$

$\lambda = 1$

$\lambda = 1$

mean $= 1$

Average No. of → Poisson always Positive



Topic : Probability and statistics

H.W

#Q. If a random variable X satisfies the poisson's distribution with a mean value of 2, then the probability that $x > 2$ is.

A $2e^{-2}$

B $1 - 2e^{-2}$

C $3e^{-2}$

D $1 - 3e^{-2}$



Topic : Probability and statistics

#Q. A traffic office imposes on an average 5 number of penalities daily on different days is independent and follows a poisson distribution.
The probability that there will be less than 4 penalties in a day is 0.26.

$\lambda = 5$

$P(X<4) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$= 0.26$

$P(X=r) = \frac{e^{-\lambda}\lambda^r}{r!}$

$P(X=0) = \frac{e^{-5}(5)^0}{0!}$

$P(X=1) = \frac{e^{-5}(5)^1}{1!}$

$P(X=2) = \frac{e^{-5}(5)^2}{2!}$

$P(X=3) = \frac{e^{-5}(5)^3}{3!}$

$P(X<4) = e^{-5} + 5e^{-5} + \frac{25e^{-5}}{2} + \frac{125}{6}e^{-5} = 0.26$



Topic : Probability and statistics

55 minutes

240 veh/hr

#Q. An observer counts 240 veh/h at a specific highway location. Assume that the vehicle arrival at the location is poisson distributed, the probability of having one vehicle arriving over a 30-second time interval is

Poisson distribution

$\lambda = 240$ veh/hr

$= \frac{240}{60}$ veh/minute

$= 4$ veh/minute

$\lambda = 2$ veh/30sec

$r = 1$

Using Poisson distribution $P(X=r) = \frac{e^{-\lambda}\lambda^r}{r!}$

$P(X=1) = \frac{e^{-2}(2)^1}{1!}$

$= 2e^{-2}$

$= Ans = 0.27$



Topic : Probability and statistics

H.W

group

Rahul Sir PW

#Q. The average number of traffic accidents on a certain section of highway is two per week assume that the number of accidents follows a poisson distribution.

1. Find the probability of no accidents on this section of highway during a 1-week period.
2. Find the probability of at most three accidents on this section of highway during a 1-week.
2. Find the probability of at least four accidents during a 1-week.
3. Find the probability of at least four accidents during a 1-week
4. Find variance and standard deviation.



Topic : Probability and statistics

13 | 14 | 15

#Q. If a random variable X has a poisson distribution with mean 5, then the expectation $E[(X+2)^2]$ equals_____.

$\lambda = 5$

$E[X^2] - [E[X]]^2 = V(X)$

$E[X^2] = \lambda + \lambda^2$

$E[(X+2)^2] = E[X^2 + 4X + 4]$

$= E[X^2] + 4E[X] + 4$

$= (\lambda + \lambda^2) + 4 \times 5 + 4$

$= 5 + 5^2 + 20 + 4$

$5 + 25 + 24$

$E[(X+2)^2] = 54$

Ratio / Proportion

PSC

THANK - YOU